



كلية تكنولوجيا الصناعة والطاقة

Information Technology Department



جامعة برج العرب التكنولوجية
BORG AL ARAB TECHNOLOGICAL UNIVERSITY

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INTRODUCTION TO C PROGRAMMING ESSENTIALS

First Year – Information Technology Program

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LECTURE 10

C LANGUAGE

C MULTIDIMENSIONAL ARRAYS



C MULTIDIMENSIONAL ARRAYS

- if you want to store data as a tabular form, like a table with rows and columns, you need to get familiar with **multidimensional arrays**.
 - Arrays can have any number of dimensions. In this lecture, we will introduce the most common; two-dimensional arrays (2D).
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- A 2D array is also known as a matrix (a table of rows and columns).

	COLUMN 0	COLUMN 1	COLUMN 2
ROW 0	1	4	2
ROW 1	3	6	8

- To create a 2D array of integers, take a look at the following example:

■ `int matrix[2][3] = { {1, 4, 2}, {3, 6, 8} };`

	COLUMN 0	COLUMN 1	COLUMN 2
ROW 0	1	4	2
ROW 1	3	6	8

- The first dimension represents the number of rows **[2]**, while the second dimension represents the number of columns **[3]**.

ACCESS THE ELEMENTS OF A 2D ARRAY

```
int matrix[2][3] = { {1, 4, 2}, {3, 6, 8} };
```

```
printf("%d", matrix[0][2]);
```

CHANGE ELEMENTS IN A 2D ARRAY

```
#include <stdio.h>
```

```
int main() {  
    int matrix[2][3] = { {1, 4, 2}, {3, 6, 8} };  
    matrix[0][0] = 9;  
    printf("%d", matrix[0][0]);  
    return 0;  
}
```

LOOP THROUGH A 2D ARRAY

- To loop through a multi-dimensional array, you need one loop for each of the array's dimensions.

```
#include <stdio.h>

int main() {
    int matrix[2][3] = { {1, 4, 2}, {3, 6, 8} };
    int i, j;
    for (i = 0; i < 2; i++) {
        for (j = 0; j < 3; j++) {
            printf("%d\n", matrix[i][j]);
        }
    }
    return 0;
}
```


- What does the following declaration represent?

```
int matrix[2][3] = { {1, 4, 2}, {3, 6, 8} };
```

- - A 1-dimensional array of 5 elements
- A 2-dimensional array with 2 rows and 3 columns
- A 2-dimensional array with 3 rows and 2 columns
- A 1-dimensional array with 6 elements

```
matrix[1][0]  
matrix[2][0]  
matrix[0][1]  
matrix[1][1]
```

- How would you access the element in the second row and first column of this array?
- `int matrix[2][3] = { {1, 4, 2}, {3, 6, 8} };`
- `matrix[1][0]`
- `matrix[2][0]`
- `matrix[0][1]`
- `matrix[1][1]`

```
int matrix[2][3] = { {1, 4, 2}, {3, 6, 8} };  
printf("%d", matrix[0][2]);  
4  
6  
2  
3
```

■ What would the following code output?

```
int matrix[2][3] = { {1, 4, 2}, {3, 6, 8} };  
printf("%d", matrix[0][2]);
```

■ 4

■ 6

■ 2

■ 3

- Which of the following statements will change the first element in the second row of matrix to 7?

```
int matrix[2][3] = { {1, 4, 2}, {3, 6, 8} };
```

- `matrix[0][1] = 7;`
- `matrix[1][0] = 7;`
- `matrix[2][0] = 7;`
- `matrix[1][1] = 7;`

To print all elements in a 2D array, what is required?

- Only one for loop
- A nested loop with one for loop for each dimension
- Two separate for loops for each dimension
- A single printf statement

MULTIPLICATION TABLE EXAMPLE

```
#include <stdio.h>
```

```
int main() {
```

```
    int i, j;
```

```
    for (i = 1; i <= 3; i++) {
```

```
        for (j = 1; j <= 3; j++) {
```

```
            printf("%d ", i * j);
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

```
}
```



Happy Coding!